

Retrieval Augmented Generation

Retrieval Augmented Generation, or RAG, combines information retrieval with large language models.

A RAG system retrieves relevant documents from an external knowledge source and provides them as context to a language model.

A typical RAG pipeline includes document loading, text splitting, embedding generation, vector storage, retrieval, prompt construction, and response generation.

BM25 is a keyword-based retrieval method. Dense retrieval uses vector embeddings to identify semantically similar documents.

BM25 works well when important query keywords occur in documents. Dense retrieval can find relevant information even when different words are used for the same meaning.

RAG evaluation can include precision, recall, MRR, nDCG, faithfulness, and answer correctness.